

Model Weights Information

Pretrained Model (No Download Required)

The AI Fitness Pose Analyzer uses **Google's MediaPipe Pose**, which is a fully pretrained model. The weights are automatically downloaded and cached by the MediaPipe Python package on first run.

- **Model:** MediaPipe Pose Landmark Model (BlazePose)
- **Source:** Google MediaPipe framework
- **Package:** mediapipe==0.10.14 (listed in requirements.txt)
- **Download:** Automatic via mediapipe pip package
- **Size:** ~25MB (downloaded to ~/.mediapipe/ on first execution)

No Custom Training Required

The pose estimation component requires:

- No custom dataset collection
- No model fine-tuning
- No separate weight file download

LLM Component (API-Based)

The AI feedback component uses **Groq API** with Llama 3.3 70B:

- No local model weights needed
- Accessed remotely via API key
- Free tier: 30 requests/minute, 14,400 requests/day

To Run

```
pip install -r requirements.txt
# MediaPipe weights download automatically on first pose detection
python inference.py --image sample_pose.jpg
```